

SQL Lab-23

Implement Stored Procedures for Reusable SQL Operations

From the table STUDENT perform the following queries:

Part – A:

1. INSERT Procedures: Create stored procedures to insert records into STUDENT tables (SP_INSERT_STUDENT).

```
CREATE PROCEDURE SP_INSERT_STUDENT(  
    p_stid INT,  
    p_sname VARCHAR(50),  
    p_city VARCHAR(50),  
    p_spi DECIMAL(4,2),  
    p_branch VARCHAR(50)  
)  
  
BEGIN  
  
    INSERT INTO STUDENT (STID, SNAME, CITY, SPI, BRANCH)  
    VALUES (p_stid, p_sname, p_city, p_spi, p_branch);  
  
END
```

2. INSERT Procedures: Create stored procedures to insert records into DEPOSIT tables (SP_INSERT_DEPOSIT).

```
CREATE PROCEDURE SP_INSERT_DEPOSIT(  
    p_actno INT,  
    p_cname VARCHAR(50),  
    p_bname VARCHAR(50),  
    p_amount DECIMAL(10,2),  
    p_adata DATE  
)  
BEGIN  
    INSERT INTO DEPOSIT (ACTNO, CNAME, BNAME, AMOUNT, ADATE)  
    VALUES (p_actno, p_cname, p_bname, p_amount, p_adata);  
END
```

3. UPDATE Procedures: Create stored procedure SP_UPDATE_STUDENT to update Branch in STUDENT table. (Update using studentID).

```
CREATE PROCEDURE SP_UPDATE_STUDENT(  
    p_stid INT,  
    p_branch VARCHAR(50)  
)  
BEGIN  
    UPDATE STUDENT  
    SET BRANCH = p_branch  
    WHERE STID = p_stid;  
END
```

4. DELETE Procedures: Create stored procedure SP_DELETE_STUDENT to delete records from STUDENT where Student Name is RAVI.

```
CREATE PROCEDURE SP_DELETE_STUDENT(  
    p_sname VARCHAR(50)  
)  
BEGIN  
    DELETE FROM STUDENT  
    WHERE SNAME = p_sname;  
END
```

5. SELECT BY PRIMARY KEY: Create stored procedures to select records by primary key (SP_SELECT_STUDENT_BY_ID) from Student table. (Display All Columns).

```
CREATE PROCEDURE SP_SELECT_STUDENT_BY_ID(  
    p_stid INT  
)  
BEGIN  
    SELECT *  
    FROM STUDENT  
    WHERE STID = p_stid;  
END
```

6. Create a stored procedure that shows details of the first 5 students ordered by SPI (Highest First).

```
CREATE PROCEDURE SP_TOP5_STUDENTS_BY_SPI()  
BEGIN  
    SELECT TOP 5 *  
    FROM STUDENT  
    ORDER BY SPI DESC  
END
```

Part – B:

7. Create a stored procedure which displays all employee details.

```
CREATE PROCEDURE SP_ALL_EMPLOYEES()
BEGIN
    SELECT * FROM EMPLOYEE;
END
```

8. Create a stored procedure that takes department name as input and returns all the employee in that department.

```
CREATE PROCEDURE SP_EMPLOYEES_BY_DEPT(
    p_deptname VARCHAR(50)
)
BEGIN
    SELECT *
    FROM EMPLOYEE
    WHERE DEPTNAME = p_deptname;
END
```

Part – C:

9. Create a stored procedure which displays department-wise maximum, minimum, and average salary of employee.

```
CREATE PROCEDURE SP_DEPT_SALARY_STATS()
BEGIN
    SELECT DEPTNAME,
           MAX(SALARY) AS MAX_SALARY,
           MIN(SALARY) AS MIN_SALARY,
           AVG(SALARY) AS AVG_SALARY
    FROM EMPLOYEE
    GROUP BY DEPTNAME;
END
```

10. Create a stored procedure that accepts department name as parameter and returns total salary of their department.

```
CREATE PROCEDURE SP_DEPT_TOTAL_SALARY(
    p_deptname VARCHAR(50)
)
BEGIN
    SELECT SUM(SALARY) AS TOTAL_SALARY
    FROM EMPLOYEE
    WHERE DEPTNAME = p_deptname;
END
```